

02

Matrices

DEFINITION, SPECIAL TYPE OF MATRICES:

Introduction:

A rectangular array of mn numbers (which may be **real or complex**) in the form of ' m ' horizontal lines (called **rows**) and ' n ' vertical lines (called **columns**), is called a matrix of order m by n , written as $m \times n$ matrix.

Such an array is enclosed by $[]$ or $()$ or $||$. An $m \times n$ matrix is usually written as

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix}$$

In compact form, the above matrix is represented by $A = [a_{ij}]_{m \times n}$. The number $a_{11}, a_{12}, \dots$ etc. are known as the elements of the matrix A , a_{ij} belongs to the i^{th} row and j^{th} column and is called the $(i, j)^{th}$ element of the matrix $A = [a_{ij}]$.

e.g. $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & 9 \end{bmatrix}$ is a matrix having 2 rows and 3 columns. Its order is 2×3 and it has 6 elements:

$$a_{11} = 1, a_{12} = 2, a_{13} = 3, a_{21} = 0, a_{22} = -1, a_{23} = 9.$$

Special Type of Matrices:

(a) **Row Matrix (Row vector)** : $A = [a_{11}, a_{12}, \dots, a_{1n}]$ i.e. row matrix has exactly one row.

(b) **Column Matrix (Column vector)** : $A = \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix}$ i.e. column matrix has exactly one column.

(c) **Zero or Null Matrix** : ($A = O_{m \times n}$) An $m \times n$ matrix whose all entries are zero.

$$A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \text{ is a } 3 \times 2 \text{ null matrix \& } B = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \text{ is } 3 \times 3 \text{ null matrix}$$

(d) **Horizontal Matrix** : A matrix of order $m \times n$ is a horizontal matrix if $n > m$ e.g. $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 5 & 1 & 1 \end{bmatrix}$

(e) **Vertical Matrix** : A matrix of order $m \times n$ is a vertical matrix if $m > n$ e.g. $\begin{bmatrix} 2 & 5 \\ 1 & 1 \\ 3 & 6 \\ 2 & 4 \end{bmatrix}$

(f) **Square Matrix** : If number of rows = number of columns $\Rightarrow$ matrix is a square matrix.

If number of rows = number of columns = n then, matrix is of the order ' n '.

Note: The pair of elements a_{ij} & a_{ji} are called **Conjugate Elements**.

Trace of Matrix:

The sum of the elements of a **square matrix** A lying along the principal diagonal is called the trace of

A i.e. ($tr(A)$). Thus, if $A = [a_{ij}]_{n \times n}$, then $tr(A) = \sum_{i=1}^n a_{ii} = a_{11} + a_{22} + \dots + a_{nn}$

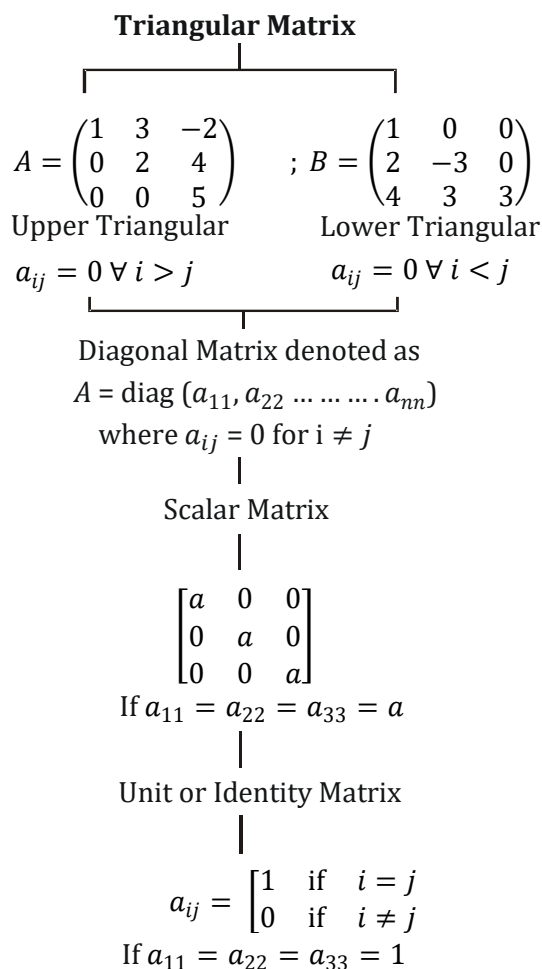
Properties of Trace of a Matrix:

Let $A = [a_{ij}]_{n \times n}$ and $B = [b_{ij}]_{n \times n}$ and λ be a scalar then

$$(i) \quad tr(\lambda A) = \lambda tr(A)$$

$$(ii) \quad tr(A + B) = tr(A) + tr(B)$$

Square Matrices:



Note:

(i) Minimum number of zeros in triangular matrix of order $n = \frac{n(n-1)}{2}$.

(ii) Minimum number of zero in a diagonal matrix of order $n = n(n-1)$.

Matrices

Illustration 1

Choose the correct answer

- (A) Every identity matrix is a scalar matrix
- (B) Every scalar matrix is an identity matrix
- (C) Every diagonal matrix is an identity matrix
- (D) A square matrix whose each element is 1 is an identity matrix

Ans. (A)

Solution:

We know that every identity matrix is a scalar matrix.

Illustration 2

$A = [a_{ij}]_{m \times n}$ is a square matrix, if

- (A) $m < n$
- (B) $m > n$
- (C) $m = n$
- (D) None of these

Ans. (C)

Solution:

For a square matrix number of columns = number of rows.

Illustration 3

In the matrix $A = \begin{bmatrix} 2 & 5 & 19 & 0 \\ 1 & 2 & 0 & 1 \\ 2 & 7 & \sqrt{3} & \sqrt{5} \end{bmatrix}$

- (i) The order of the matrix,
- (ii) The number of elements,
- (iii) Write the elements $a_{13}, a_{21}, a_{33}, a_{24}, a_{23}$.

Solution:

- (i) 3×4
- (ii) 12
- (iii) 19, 1, $\sqrt{3}$, 1, 0

ALGEBRA OF MATRICES:

Equality of Matrices:

Let $A = [a_{ij}]$ & $B = [b_{ij}]$ are equal if

- (a) both have the same order.
- (b) $a_{ij} = b_{ij}$ for each pair of i & j .

Illustration 4

Find the value of x, y, z and w which satisfy the matrix equation $\begin{bmatrix} x+3 & 2y+x \\ z-1 & 4w-8 \end{bmatrix} = \begin{bmatrix} -x-1 & 0 \\ 3 & 2w \end{bmatrix}$.

Solution:

As the given matrices are equal so their corresponding elements are equal.

$$\Rightarrow x + 3 = -x - 1 \Rightarrow 2x = -4$$

$$\therefore x = -2 \quad \dots(i)$$

$$\Rightarrow 2y + x = 0 \Rightarrow 2y - 2 = 0 \quad \text{[from (i)]}$$

$$\Rightarrow y = 1 \quad \dots(ii)$$

$$\Rightarrow z - 1 = 3 \Rightarrow z = 4 \quad \dots(iii)$$

$$\Rightarrow 4w - 8 = 2w \Rightarrow 2w = 8$$

$$\therefore w = 4 \quad \dots(iv)$$

Addition: $A + B = [a_{ij} + b_{ij}]$ where A & B are of the same order.

(a) **Addition of matrices is commutative:** i.e. $A + B = B + A$

(b) **Matrix addition is associative:** $(A + B) + C = A + (B + C)$

(c) **Additive inverse:** If $A + B = O = B + A$, then B is called **additive inverse** of A .

(d) **Existence of additive identity:** Let $A = [a_{ij}]$ be an $m \times n$ matrix and O be an $m \times n$ zero matrix, then $A + O = O + A = A$. In other words, O is the **additive identity** for matrix addition.

(e) **Cancellation laws** hold good in case of addition of matrices. If A, B, C are matrices of the same order, then $A + B = A + C \Rightarrow B = C$ (**left cancellation law**) and $B + A = C + A \Rightarrow B = C$ (**right-cancellation law**)

Note: The zero matrix plays the same role in matrix addition as the number zero does in addition of numbers.

Illustration 5

If $A = \begin{bmatrix} 1 & 3 \\ 3 & 2 \\ 2 & 5 \end{bmatrix}$ & $B = \begin{bmatrix} -1 & -2 \\ 0 & 5 \\ 3 & 1 \end{bmatrix}$ and $A + B - D = O$ (zero matrix), then D matrix will be-

(A) $\begin{bmatrix} 0 & 2 \\ 3 & 7 \\ 6 & 5 \end{bmatrix}$

(B) $\begin{bmatrix} 0 & 2 \\ 3 & 7 \\ 5 & 6 \end{bmatrix}$

(C) $\begin{bmatrix} 0 & 1 \\ 3 & 7 \\ 5 & 6 \end{bmatrix}$

(D) $\begin{bmatrix} 0 & -2 \\ -3 & -7 \\ -5 & -6 \end{bmatrix}$

Ans. (C)

Solution:

Let $D = \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix}$

$$\therefore A + B - D = \begin{bmatrix} 1 & 3 \\ 3 & 2 \\ 2 & 5 \end{bmatrix} + \begin{bmatrix} -1 & -2 \\ 0 & 5 \\ 3 & 1 \end{bmatrix} - \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix} \Rightarrow \begin{bmatrix} 1-1-a & 3-2-b \\ 3+0-c & 2+5-d \\ 2+3-e & 5+1-f \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow -a = 0 \Rightarrow a = 0, 1-b = 0 \Rightarrow b = 1,$$

$$\Rightarrow 3-c = 0 \Rightarrow c = 3, 7-d = 0 \Rightarrow d = 7,$$

$$\Rightarrow 5-e = 0 \Rightarrow e = 5, 6-f = 0 \Rightarrow f = 6$$

$$\therefore D = \begin{bmatrix} 0 & 1 \\ 3 & 7 \\ 5 & 6 \end{bmatrix}$$

Multiplication of a Matrix by a Scalar:

If $A = \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}$; $kA = \begin{bmatrix} ka & kb & kc \\ kb & kc & ka \\ kc & ka & kb \end{bmatrix}$

Properties of Scalar Multiplication:

(a) If A and B are two matrices of the same order and ' k ' be a scalar then $k(A + B) = kA + kB$.

(b) If k_1 and k_2 are two scalars and ' A ' is a matrix, then $(k_1 + k_2)A = k_1A + k_2A$.

(c) If k_1 and k_2 are two scalars and ' A ' is a matrix, then $(k_1k_2)A = k_1(k_2A) = k_2(k_1A)$

Multiplication of Matrices (Row by Column):

Let A be a matrix of order $m \times n$ and B be a matrix of order $p \times q$, then the matrix multiplication AB is possible if and only if $n = p$ and matrices are said to be **conformable** for multiplication.

In the product AB , A is called pre-factor and B is called post factor.

$\Rightarrow AB$ is possible if and only if number of columns in pre-factor = number of rows in post-factor.

Let $A_{m \times n} = [a_{ij}]$ and $B_{n \times p} = [b_{ij}]$, then order of AB is $m \times p$ & $(AB)_{ij} = \sum_{r=1}^n a_{ir} b_{rj}$

e.g. $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}_{2 \times 3}$ and $B = \begin{bmatrix} b_{11} & b_{12} & b_{13} & b_{14} \\ b_{21} & b_{22} & b_{23} & b_{24} \\ b_{31} & b_{32} & b_{33} & b_{34} \end{bmatrix}_{3 \times 4}$

Then order of AB is 2×4 .

$$(AB)_{11} = a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} = \sum_{r=1}^3 a_{1r}b_{r1}$$

$$(AB)_{23} = a_{21}b_{13} + a_{22}b_{23} + a_{23}b_{33} = \sum_{r=1}^3 a_{2r}b_{r3}$$

In general, $(AB)_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + a_{i3}b_{3j} = \sum_{r=1}^3 a_{ir}b_{rj}$

Illustration 6

If $\begin{bmatrix} 1 & x & 2 \end{bmatrix} \begin{bmatrix} 2 & 3 & 1 \\ 0 & 4 & 2 \\ 0 & 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ 1 \\ -1 \end{bmatrix} = 0$, then the value of x is

- (A) -1 (B) 0 (C) 1 (D) 2

Ans. (A)

Solution:

The LHS of the equation

$$= [24x + 92x + 5] \begin{bmatrix} x \\ 1 \\ -1 \end{bmatrix} = [2x + 4x + 9 - 2x - 5] = 4x + 4$$

Thus $4x + 4 = 0$; $x = -1$

Illustration 7

If A, B are two matrices such that $A + B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$, $A - B = \begin{bmatrix} 3 & 2 \\ -2 & 0 \end{bmatrix}$, then find AB .

Solution:

Given $A + B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$... (i) &

$\Rightarrow A - B = \begin{bmatrix} 3 & 2 \\ -2 & 0 \end{bmatrix}$... (ii)

Adding (i) & (ii)

$$\Rightarrow 2A = \begin{bmatrix} 4 & 4 \\ 0 & 4 \end{bmatrix} \Rightarrow A = \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix}$$

Subtracting (ii) from (i)

$$\Rightarrow 2B = \begin{bmatrix} -2 & 0 \\ 4 & 4 \end{bmatrix} \Rightarrow B = \begin{bmatrix} -1 & 0 \\ 2 & 2 \end{bmatrix}$$

$$\text{Now } AB = \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 4 & 4 \end{bmatrix}$$

PROPERTIES OF MATRIX MULTIPLICATION:

(a) Matrix Multiplication is not Commutative: i.e. $AB \neq BA$

Here both AB & BA exist and also, they are of the same type but $AB \neq BA$.

$$\text{e.g. Let } A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \text{ \& } B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix};$$

$$\text{then } AB = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}; \quad BA = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

$AB \neq BA$ (in general)

(b) $AB = \mathbf{O} \nRightarrow A = \mathbf{O}$ or $B = \mathbf{O}$ (in general)

$$\text{e.g. Let } A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} \text{ \& } B = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}, \text{ then } AB = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Note:

If A and B are two non-zero matrices such that $AB = \mathbf{O}$ then A and B are called the divisors of zero.

If A and B are two matrices such that

(i) $AB = BA$ then A and B are said to commute

(ii) $AB = -BA$ then A and B are said to anti-commute

(c) Matrix Multiplication Is Associative:

If A, B & C are conformable for the product AB & BC , then $(AB)C = A(BC)$

(d) Distributivity:

$$\begin{aligned} A(B+C) &= AB+AC \\ (A+B)C &= AC+BC \end{aligned} \quad \text{Provided } A, B \text{ \& } C \text{ are conformable for respective products}$$

(e) $\text{tr}(AB) = \text{tr}(BA)$

Illustration 8

$$\text{Let } A = \begin{bmatrix} 1 & -3 & 2 \\ 2 & 1 & -3 \\ 4 & -3 & -1 \end{bmatrix}, B = \begin{bmatrix} 1 & 4 & 1 & 0 \\ 2 & 1 & 1 & 1 \\ 1 & -2 & 1 & 2 \end{bmatrix} \text{ \& } C = \begin{bmatrix} 1 & 1 & -1 & -2 \\ 3 & -2 & -1 & -1 \\ 2 & -5 & -1 & 0 \end{bmatrix} \text{ be the matrices then, prove that in matrix}$$

multiplication cancellation law does not hold.

Solution:

We have to show that $AB = AC$; though B is not equal to C .

$$\text{We have } AB = \begin{bmatrix} 1 & -3 & 2 \\ 2 & 1 & -3 \\ 4 & -3 & -1 \end{bmatrix} \begin{bmatrix} 1 & 4 & 1 & 0 \\ 2 & 1 & 1 & 1 \\ 1 & -2 & 1 & 2 \end{bmatrix} = \begin{bmatrix} -3 & -3 & 0 & 1 \\ 1 & 15 & 0 & -5 \\ -3 & 15 & 0 & -5 \end{bmatrix}_{3 \times 4}$$

$$\text{Now, } AC = \begin{bmatrix} 1 & -3 & 2 \\ 2 & 1 & -3 \\ 4 & -3 & -1 \end{bmatrix} \begin{bmatrix} 2 & 1 & -1 & -2 \\ 3 & -2 & -1 & -1 \\ 2 & -5 & -1 & 0 \end{bmatrix} = \begin{bmatrix} -3 & -3 & 0 & 1 \\ 1 & 15 & 0 & -5 \\ -3 & 15 & 0 & -5 \end{bmatrix}_{3 \times 4}$$

Here, $AB = AC$ though B is not equal to C . Thus, cancellation law does not hold in general.

Illustration 9

If $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & -1 & 3 \\ -1 & 0 & 2 \end{bmatrix}$, then find $2A - B$.

Solution:

We have

$$\begin{aligned} 2A - B &= 2 \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix} - \begin{bmatrix} 3 & -1 & 3 \\ -1 & 0 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 2 & 4 & 6 \\ 4 & 6 & 2 \end{bmatrix} + \begin{bmatrix} -3 & -1 & -3 \\ 1 & 0 & -2 \end{bmatrix} \\ &= \begin{bmatrix} -1 & 5 & 3 \\ 5 & 6 & 0 \end{bmatrix} \end{aligned}$$

Illustration 10

Which of the following holds true for matrix multiplication?

- (A) follows commutative property (B) follows distributive property
(C) follows associative property (D) None of these

Ans. (B) and (C)

POSITIVE INTEGRAL POWERS OF A SQUARE MATRIX:

For a square matrix A , $A^n = \underbrace{A.A.A \dots A}_{\text{upto } n \text{ times}}$, where $n \in N$

Note:

- (i) $A^m \cdot A^n = A^{m+n}$
(ii) $(A^m)^n = A^{mn}$, where $m, n \in N$
(iii) If A and B are square matrices of same order and $AB = BA$ then
 $(A + B)^n = {}^nC_0 A^n + {}^nC_1 A^{n-1} B + {}^nC_2 A^{n-2} B^2 + \dots + {}^nC_n B^n$

Note that for a unit matrix I of any order, $I^m = I$ for all $m \in N$.

Illustration 11

If $A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$, then $A^5 =$

- (A) $5A$ (B) $10A$ (C) $16A$ (D) $32A$

Ans. (C)

Solution:

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} \Rightarrow A^5 = \begin{bmatrix} 2^5 & 0 & 0 \\ 0 & 2^5 & 0 \\ 0 & 0 & 2^5 \end{bmatrix} = 2^4 \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} = 16A.$$

Illustration 12

A square matrix P satisfies $P^2 = I - P$, where I is the identity matrix. If $P^n = 5I - 8P$ ($n \in N$), then minimum value of n is equal to

- (A) 4 (B) 5 (C) 6 (D) 7

Ans. (C)

Solution:

Since $P^2 = I - P$

$$\Rightarrow P^3 = P(I - P) = P - P^2 = P - (I - P) = 2P - I$$

Similarly, $P^4 = 2P^2 - P = 2I - 3P$ and

$$\Rightarrow P^5 = 5P - 3I$$

$$\Rightarrow P^6 = 5P^2 - 3P = 5I - 8P$$

So $n = 6$.

Illustration 13

If $A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$, show that $A^k = \begin{bmatrix} 1+2k & -4k \\ k & 1-2k \end{bmatrix}$, where k is any positive integer.

Solution:

We have,

$$\Rightarrow A^2 = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 5 & -8 \\ 2 & -3 \end{bmatrix} = \begin{bmatrix} 1+2 \times 2 & -4 \times 2 \\ 2 & 1-2 \times 2 \end{bmatrix} \text{ and}$$

$$\Rightarrow A^3 = \begin{bmatrix} 5 & -8 \\ 2 & -3 \end{bmatrix} \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 7 & -12 \\ 3 & -5 \end{bmatrix} = \begin{bmatrix} 1+2 \times 3 & -4 \times 3 \\ 3 & 1-2 \times 3 \end{bmatrix}$$

Thus, it is true for indices 2 and 3. Now assume

$$\Rightarrow A^k = \begin{bmatrix} 1+2k & -4k \\ k & 1-2k \end{bmatrix}$$

$$\text{Then, } A^{k+1} = \begin{bmatrix} 1+2k & -4k \\ k & 1-2k \end{bmatrix} \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 3+2k & -4(k+1) \\ k+1 & -1-2k \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1+2(k+1) & -4(k+1) \\ k+1 & 1-2(k+1) \end{bmatrix}$$

Thus, if the law is true for A^k , it is also true for A^{k+1} . But it is true for $k = 2, 3$ etc. Hence, by induction, the required result follows.

THE TRANSPOSE OF A MATRIX: (CHANGING ROWS & COLUMNS)

Let A be any matrix of order $m \times n$. Then A^T or $A' = [a_{ji}]$ for $1 \leq i \leq m$ & $1 \leq j \leq n$ of order $n \times m$

Properties of Transpose:

If A^T & B^T denote the transpose of A and B ,

(a) $(A + B)^T = A^T + B^T$; note that A and B have the same order.

(b) $(AB)^T = B^T A^T$ (Reversal law) A and B are conformable for matrix product AB

Note: In general: $(A_1 \cdot A_2 \cdot \dots \cdot A_n)^T = A_n^T \cdot \dots \cdot A_2^T \cdot A_1^T$ (reversal law for transpose)

(c) $(A^T)^T = A$

(d) $(kA)^T = kA^T$, k is a scalar.

Illustration 14

If A and B are matrices of order $m \times n$ and $n \times m$ respectively, then order of matrix $B^T(A^T)^T$ is -

- (A) $m \times n$ (B) $m \times m$ (C) $n \times n$ (D) Not defined

Ans. (D)

Solution:

Order of B is $n \times m$ so order of B^T will be $m \times n$

Now $(A^T)^T = A$ & its order is $m \times n$. For the multiplication $B^T(A^T)^T$

Number of columns in pre-factor $\neq$ Number of rows in post-factor.

Hence this multiplication is not defined.

Illustration 15

$$A = \begin{bmatrix} 2 & -1 \\ -7 & 4 \end{bmatrix} \text{ and } B = \begin{bmatrix} 4 & 1 \\ 7 & 2 \end{bmatrix} \text{ then } B^T A^T \text{ is}$$

(A) a null matrix

(B) an identity matrix

(C) scalar, but not an identity matrix

(D) such that $\text{Tr}(B^T A^T) = 4$

Ans. (B)

Solution:

$$A = \begin{bmatrix} 2 & -1 \\ -7 & 4 \end{bmatrix}, B = \begin{bmatrix} 4 & 1 \\ 7 & 2 \end{bmatrix}$$

$$\Rightarrow A^T = \begin{bmatrix} 2 & -7 \\ -1 & 4 \end{bmatrix}, B^T = \begin{bmatrix} 4 & 7 \\ 1 & 2 \end{bmatrix}$$

$B^T A^T$ is an identity matrix.

ORTHOGONAL MATRICES:

A square matrix is said to be orthogonal matrix if $AA^T = I$

Note:

(i) The determinant value of orthogonal matrix is either 1 or -1 .

$$(ii) \text{ Let } A = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix} \Rightarrow A^T = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$$

$$AA^T = \begin{bmatrix} a_1^2 + a_2^2 + a_3^2 & a_1b_1 + a_2b_2 + a_3b_3 & a_1c_1 + a_2c_2 + a_3c_3 \\ b_1a_1 + b_2a_2 + b_3a_3 & b_1^2 + b_2^2 + b_3^2 & b_1c_1 + b_2c_2 + b_3c_3 \\ c_1a_1 + c_2a_2 + c_3a_3 & c_1b_1 + c_2b_2 + c_3b_3 & c_1^2 + c_2^2 + c_3^2 \end{bmatrix}$$

If $AA^T = I$, then

$$\sum_{i=1}^3 a_i^2 = \sum_{i=1}^3 b_i^2 = \sum_{i=1}^3 c_i^2 = 1 \text{ and } \sum_{i=1}^3 a_i b_i = \sum_{i=1}^3 b_i c_i = \sum_{i=1}^3 c_i a_i = 0$$

Illustration 16

Determine the values of α, β, γ when $\begin{bmatrix} 0 & 2\beta & \gamma \\ \alpha & \beta & -\gamma \\ \alpha & -\beta & \gamma \end{bmatrix}$ is orthogonal.

Solution:

$$\text{Let } A = \begin{bmatrix} 0 & 2\beta & \gamma \\ \alpha & \beta & -\gamma \\ \alpha & -\beta & \gamma \end{bmatrix}$$

$$\Rightarrow A' = \begin{bmatrix} 0 & \alpha & \alpha \\ 2\beta & \beta & -\beta \\ \gamma & -\gamma & \gamma \end{bmatrix}$$

But given A is orthogonal.

$$AA^T = I$$

$$\Rightarrow \begin{bmatrix} 0 & 2\beta & \gamma \\ \alpha & \beta & -\gamma \\ \alpha & -\beta & \gamma \end{bmatrix} \begin{bmatrix} 0 & \alpha & \alpha \\ 2\beta & \beta & -\beta \\ \gamma & -\gamma & \gamma \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 4\beta^2 + \gamma^2 & 2\beta^2 - \gamma^2 & -2\beta^2 + \gamma^2 \\ 2\beta^2 - \gamma^2 & \alpha^2 + \beta^2 + \gamma^2 & \alpha^2 - \beta^2 - \gamma^2 \\ -2\beta^2 + \gamma^2 & \alpha^2 - \beta^2 - \gamma^2 & \alpha^2 + \beta^2 + \gamma^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Equating the corresponding elements, we have

$$\Rightarrow 4\beta^2 + \gamma^2 = 1 \quad \dots(i)$$

$$\Rightarrow 2\beta^2 - \gamma^2 = 0 \quad \dots(ii)$$

$$\Rightarrow \alpha^2 + \beta^2 + \gamma^2 = 1 \quad \dots(iii)$$

$$\text{From (i) and (ii), } 6\beta^2 = 1 \quad \beta^2 = \frac{1}{6} \quad \text{and} \quad \gamma^2 = \frac{1}{3}$$

$$\text{From (iii) } \alpha^2 = 1 - \beta^2 - \gamma^2 = 1 - \frac{1}{6} - \frac{1}{3} = \frac{1}{2}$$

$$\text{Hence, } \alpha = \pm \frac{1}{\sqrt{2}}, \beta = \pm \frac{1}{\sqrt{6}} \quad \text{and} \quad \gamma = \pm \frac{1}{\sqrt{3}}.$$

Illustration 17

$$\text{Let } A = \begin{pmatrix} 0 & 2q & r \\ p & q & -r \\ p & -q & r \end{pmatrix}. \text{ If } AA^T = I^3, \text{ then } |p| \text{ is :}$$

(A) $\frac{1}{\sqrt{2}}$

(B) $\frac{1}{\sqrt{5}}$

(C) $\frac{1}{\sqrt{6}}$

(D) $\frac{1}{\sqrt{3}}$

Ans. (A)

Solution:

A is orthogonal matrix

$$\Rightarrow 0^2 + p^2 + p^2 = 1 \Rightarrow |p| = \frac{1}{\sqrt{2}}$$

SYMMETRIC & SKEW SYMMETRIC MATRIX:**(a) Symmetric Matrix:**

A square matrix $A = [a_{ij}]$ is said to be, symmetric if, $a_{ij} = a_{ji} \forall i \& j$ (conjugate elements are equal).

Hence for symmetric matrix $A = A^T$.

Note: Max. number of distinct entries in any symmetric matrix of order n is $\frac{n(n+1)}{2}$.

(b) Skew Symmetric Matrix:

Square matrix $A = [a_{ij}]$ is said to be skew symmetric if $a_{ij} = -a_{ji} \forall i \& j$ (the pair of conjugate elements are additive inverse of each other). For a skew symmetric matrix, $A = -A^T$.

Note:

(i) If A is skew symmetric, then $a_{ii} = -a_{ii} \Rightarrow a_{ii} = 0 \forall i$. Thus, the diagonal elements of a skew square matrix are all zero, but not the converse.

(ii) The determinant value of odd order skew symmetric matrix is zero.

Illustration 18

If $A = \begin{bmatrix} -2 & -1 & 1 \\ -1 & 7 & 4 \\ 1 & -x & -3 \end{bmatrix}$ be symmetric matrix then find the value of x .

Solution:

$$x = -4$$

for symmetric matrix $A = A^T$

$$\Rightarrow \begin{bmatrix} -2 & -1 & 1 \\ -1 & 7 & 4 \\ 1 & -x & -3 \end{bmatrix} = \begin{bmatrix} -2 & -1 & 1 \\ -1 & 7 & -x \\ 1 & 4 & -3 \end{bmatrix}$$

$$\therefore x = -4$$

Illustration 19

$$\text{Given matrix } A = \begin{bmatrix} 1 & x & 1 \\ x & 2 & y \\ 1 & y & 3 \end{bmatrix}; B = \begin{bmatrix} 3 & -3 & z \\ -3 & 2 & -3 \\ z & -3 & 1 \end{bmatrix}.$$

Obtain x, y, z if the matrix AB is symmetric.

Solution:

$$AB = \begin{bmatrix} 1 & x & 1 \\ x & 2 & y \\ 1 & y & 3 \end{bmatrix} \begin{bmatrix} 3 & -3 & z \\ -3 & 2 & -3 \\ z & -3 & 1 \end{bmatrix} = \begin{bmatrix} 3-3x+z & -3+2x-3 & z-3x+1 \\ 3x-6+yz & -3x+4-3y & xz-6+y \\ 3-3y+3z & -3+2y-9 & z-3y+3 \end{bmatrix}$$

AB is symmetric matrix,

$$\text{So, } -6+2x = 3x-6+yz \quad \dots(i)$$

$$\Rightarrow 3-3y+3z = z-3x+1 \quad \dots(ii)$$

$$\Rightarrow -3+2y-9 = xz-6+y \quad \dots(iii)$$

So possible set of values of x, y, z

$$\Rightarrow \left(-\frac{4\sqrt{2}}{3}, \frac{2}{3}, 2\sqrt{2}\right), \left(\frac{4\sqrt{2}}{3}, \frac{2}{3}, -2\sqrt{2}\right), (3, 3, -1)$$

PROPERTIES OF SYMMETRIC & SKEW SYMMETRIC MATRIX:

- (i) A is symmetric if $A^T = A$ and A is skew symmetric if $A^T = -A$
- (ii) Let A be any square matrix then, $A + A^T$ is a symmetric matrix and $A - A^T$ is a skew symmetric matrix.
- (iii) The sum of two symmetric matrix is a symmetric matrix and the sum of two skew symmetric matrix is a skew symmetric matrix.
- (iv) If A and B are symmetric matrices then,
 - (1) $AB + BA$ is a symmetric matrix
 - (2) $AB - BA$ is a skew symmetric matrix.
- (v) Every square matrix can be uniquely expressed as a sum or difference of a symmetric and a skew symmetric matrix.

$$A = \underbrace{\frac{1}{2}(A + A^T)}_{\text{symmetric}} + \underbrace{\frac{1}{2}(A - A^T)}_{\text{skew symmetric}} \quad \text{and} \quad A = \frac{1}{2}(A^T + A) - \frac{1}{2}(A^T - A)$$

Illustration 20

If A is symmetric as well as skew symmetric matrix, then A is -

- (A) diagonal matrix
- (B) null matrix
- (C) triangular matrix
- (D) none of these

Ans. (B)

Solution:

Let $A = [a_{ij}]$ Since A is skew symmetric $a_{ij} = -a_{ji}$

for $i = j$, $a_{ii} = -a_{ii} \Rightarrow a_{ii} = 0$

for $i \neq j$, $a_{ij} = -a_{ji}$ [$\because A$ is skew symmetric] and $a_{ij} = a_{ji}$ [$\because A$ is symmetric]

$\therefore a_{ij} = 0$ for all $i \neq j$

So, $a_{ij} = 0$ for all ' i ' and ' j ' i.e. A is null matrix.

Illustration 21

Which one of the following is wrong?

- (A) The elements on the main diagonal of a symmetric matrix are all zero
- (B) The elements on the main diagonal of a skew - symmetric matrix are all zero
- (C) For any square matrix A , $\frac{1}{2}(A + A')$ is symmetric matrix
- (D) For any square matrix A , $\frac{1}{2}(A - A')$ is skew - symmetric matrix

Ans. (A)

Solution:

The elements of main diagonal of skew symmetric matrix are all zero but not necessarily for symmetric matrix.

$\Rightarrow \frac{A + A'}{2}$ is symmetric matrix.

$\Rightarrow \frac{A - A'}{2}$ is skew symmetric matrix.

Matrices

Illustration 22

Let $A = [a_{ij}]_{n \times n}$ where $a_{ij} = i^2 - j^2$. Then A is

- (A) null matrix (B) symmetric matrix
(C) skew-symmetric matrix (D) unit matrix

Ans. (C)

Solution:

$$A = [a_{ij}] = \begin{bmatrix} 1^2 - 1^2 & 1^2 - 2^2 & 1^2 - 3^2 \\ 2^2 - 1^2 & 2^2 - 2^2 & 2^2 - 3^2 \\ 3^2 - 1^2 & 3^2 - 2^2 & 3^2 - 3^2 \end{bmatrix} = \begin{bmatrix} 0 & -3 & -8 \\ 3 & 0 & -5 \\ 8 & 5 & 0 \end{bmatrix}$$

So A is skew symmetric matrix.

ADJOINT OF A SQUARE MATRIX:

Let $A = [a_{ij}]$ be a square matrix of order n and let C_{ij} be cofactor of a_{ij} in A then the adjoint of A , denoted by $\text{adj } A$, is defined as the transpose of the cofactor matrix.

$$\text{Then, } \text{adj } A = [C_{ij}]^T \Rightarrow \text{adj } A = \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix}^T$$

Theorem: $A(\text{adj } A) = (\text{adj } A) \cdot A = |A| I_n$.

$$\begin{aligned} \text{Proof: } A(\text{adj } A) &= \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} C_{11} & C_{21} & C_{31} \\ C_{12} & C_{22} & C_{32} \\ C_{13} & C_{23} & C_{33} \end{pmatrix} \\ &= \begin{pmatrix} |A| & 0 & 0 \\ 0 & |A| & 0 \\ 0 & 0 & |A| \end{pmatrix} = |A| \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \Rightarrow A(\text{adj } A) = |A| I \end{aligned}$$

(whatever may be the value only $|A|$ will come out as a common element)

If $|A| \neq 0$, then $\frac{A(\text{adj } A)}{|A|} = I = \text{unit matrix of the same order as that of } A$

Illustration 23

If $A = \begin{bmatrix} 1 & 3 & 5 \\ 3 & 5 & 1 \\ 5 & 1 & 3 \end{bmatrix}$, then $\text{adj } A$ is equal to -

(A) $\begin{bmatrix} 14 & -4 & -22 \\ -4 & -22 & 14 \\ -22 & 14 & -4 \end{bmatrix}$

(B) $\begin{bmatrix} -14 & 4 & 22 \\ 4 & 22 & -14 \\ 22 & -14 & 4 \end{bmatrix}$

(C) $\begin{bmatrix} 14 & 4 & -22 \\ 4 & -22 & -14 \\ -22 & -14 & -4 \end{bmatrix}$

(D) none of these

Ans. (A)

Solution:

$$\text{adj. } A = \begin{bmatrix} 14 & -4 & -22 \\ -4 & -22 & 14 \\ -22 & 14 & -4 \end{bmatrix}^T = \begin{bmatrix} 14 & -4 & -22 \\ -4 & -22 & 14 \\ -22 & 14 & -4 \end{bmatrix}$$

Illustration 24

If $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$, then $\text{adj } A =$

(A) $\begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix}$ (B) $\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$ (C) $\begin{bmatrix} 1 & -2 \\ -2 & -1 \end{bmatrix}$ (D) $\begin{bmatrix} -1 & 2 \\ 2 & -1 \end{bmatrix}$

Ans. (A)

Solution:

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$

Matrix formed by cofactors of $A = C = \begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix}$

Transpose of Matrix $C = C^T = \begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix}$

Adjoint of matrix $A = C^T = \text{adj } A = \begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix}$

Illustration 25

If $A = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$, then $\text{adj } A =$

(A) A^2 (B) I (C) O (D) A'

Ans. (D)

Solution:

$$A = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Matrix formed by Cofactors of $A = C = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$\therefore \text{Adj } A = C^T = \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} = A'$$

PROPERTIES OF ADJOINT:

If A be a square matrix of order n , then

- (i) $|adj A| = |A|^{n-1}$
- (ii) $adj(adj A) = |A|^{n-2} A$, where $|A| \neq 0$
- (iii) $|adj(adj A)| = |A|^{(n-1)^2}$, where $|A| \neq 0$
- (iv) $adj(AB) = (adj B)(adj A)$
- (v) $adj(KA) = K^{n-1}(adj A)$, K is a scalar
- (vi) $adj A^T = (adj A)^T$

Method to Find Adjoint of a 2×2 Square Matrix, Directly:

Let A be a 2×2 square matrix. In order to find the adjoint simply interchange the diagonal elements and reverse the sign of off diagonal elements (rest of the elements).

e.g. If $A = \begin{bmatrix} p & q \\ r & s \end{bmatrix} \Rightarrow adj A = \begin{bmatrix} s & -q \\ -r & p \end{bmatrix}$

Illustration 26

If $A = \begin{bmatrix} 2 & 0 & 0 \\ 2 & 2 & 0 \\ 2 & 2 & 2 \end{bmatrix}$, then $adj(adj A)$ is equal to -

- (A) $8 \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$ (B) $16 \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$ (C) $64 \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$ (D) none of these

Ans. (B)

Solution:

$$|A| = \begin{vmatrix} 2 & 0 & 0 \\ 2 & 2 & 0 \\ 2 & 2 & 2 \end{vmatrix} = 8$$

$$\text{Now } adj(adj A) = |A|^{3-2} A$$

$$= 8 \begin{bmatrix} 2 & 0 & 0 \\ 2 & 2 & 0 \\ 2 & 2 & 2 \end{bmatrix} = 16 \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

Illustration 27

If A is a square matrix of order $n \times n$ and k is a scalar, then $adj(kA)$ is equal to

- (A) $k adj A$ (B) $k^n adj A$ (C) $k^{n-1} adj A$ (D) $k^{n+1} adj A$

Ans. (C)

Solution:

$$kA adj(kA) = |kA| I_n$$

$$\Rightarrow kA adj(kA) = k^n |A| I_n$$

$$\Rightarrow kA adj(kA) = k^n A adj A$$

Pre-multiplying A^{-1}

$$\Rightarrow adj(kA) = k^{n-1} adj A$$

Illustration 28

If A is square matrix of order n then $\text{adj}(\text{adj } A) =$

- (A) $|A|^{n-1} A$ (B) $|A|^{(n-2)} A$ (C) $|A|^{n-2}$ (D) $|A|^n A$

Ans. (B)

Solution:

We know that $A \cdot \text{adj } A = |A| \cdot I$ and $|\text{adj } A| = |A|^{n-1}$

$$\therefore \text{adj } A \cdot (\text{adj adj } A) = |\text{adj } A| I$$

Pre-multiplying A

$$\Rightarrow A \cdot \text{adj } A \cdot (\text{adj adj } A) = A |A|^{n-1} I$$

$$\Rightarrow |A| \cdot I (\text{adj adj } A) = A |A|^{n-1}$$

$$\Rightarrow (\text{adj adj } A) = A |A|^{n-2}$$

Illustration 29

If A is square matrix of order 3 then $|\text{adj}(\text{adj } A)| =$

- (A) $|A|$ (B) $|A|^2$ (C) $|A|^3$ (D) $|A|^4$

Ans. (D)

Solution:

$\because A$ is square matrix of order 3

$$\therefore |\text{adj } A| = |A|^2$$

$$\therefore |\text{adj}(\text{adj } A)| = |\text{adj } A|^2 = (|A|^2)^2 = |A|^4$$

INVERSE OF A MATRIX (RECIPROCAL MATRIX):

A square matrix A said to be invertible if and only if it is non-singular (i. e. $|A| \neq 0$) and there exists a matrix B such that, $AB = I = BA$.

B is called the **inverse** (reciprocal) of A and is denoted by A^{-1} . Thus

$$A^{-1} = B \Leftrightarrow AB = I = BA$$

We have, $A \cdot (\text{adj } A) = |A| I_n$

$$A^{-1} \cdot A (\text{adj } A) = A^{-1} I_n |A|$$

$$I_n (\text{adj } A) = A^{-1} |A| I_n$$

$$\therefore A^{-1} = \frac{(\text{adj } A)}{|A|}$$

Note: The necessary and sufficient condition for a square matrix A to be invertible is that $|A| \neq 0$

PROPERTIES OF INVERSE:

(i) If A and B are invertible matrices of the same order, then $(AB)^{-1} = B^{-1}A^{-1}$.

Note: If $A_1, A_2, \dots, A_n$ are all invertible square matrices of order n

$$\text{then } (A_1 A_2 \dots A_n)^{-1} = A_n^{-1} A_{n-1}^{-1} \dots A_2^{-1} A_1^{-1}$$

(ii) If A be an invertible matrix, then A^T is also invertible and $(A^T)^{-1} = (A^{-1})^T$.

(iii) If A is invertible, (a) $(A^{-1})^{-1} = A$ (b) $(A^k)^{-1} = (A^{-1})^k = A^{-k}; k \in \mathbb{N}$

(iv) If A is non-singular matrix, then $|A^{-1}| = |A|^{-1}$

(v) If idempotent matrix is invertible then its inverse will be identity matrix.

(vi) A nilpotent matrix will not be invertible because its determinant value is zero.

(vii) Orthogonal matrix A is always invertible and $A^{-1} = A^T$.

(viii) $A = A^{-1}$ for an involutory matrix.

Cancellation Law:

Let A, B, C be square matrices of the same order ' n '.

If A is a non-singular matrix, then

(a) $AB = AC \Rightarrow B = C$ (Left cancellation law)

(b) $BA = CA \Rightarrow B = C$ (Right cancellation law)

Note that these cancellation laws hold only if the matrix ' A ' is **non – singular** (i.e. $|A| \neq 0$).

Illustration 30

Prove that if A is non – singular matrix such that A is symmetric then A^{-1} is also symmetric.

Solution:

$$A^T = A \quad [\because A \text{ is a symmetric matrix}]$$

$$\Rightarrow (A^T)^{-1} = A^{-1} \text{ [since } A \text{ is non-singular matrix]}$$

$$\Rightarrow (A^{-1})^T = A^{-1} \text{ Hence proved}$$

Illustration 31

$$\begin{bmatrix} 1 & -\tan \theta / 2 \\ \tan \theta / 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & \tan \theta / 2 \\ -\tan \theta / 2 & 1 \end{bmatrix}^{-1} \text{ is equal to -}$$

$$(A) \begin{bmatrix} \sin \theta & -\cos \theta \\ \cos \theta & \sin \theta \end{bmatrix}$$

$$(B) \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

$$(C) \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

(D) none of these

Ans. (C)

Solution:

$$\begin{bmatrix} 1 & \tan \theta / 2 \\ -\tan \theta / 2 & 1 \end{bmatrix}^{-1} = \frac{1}{\sec^2 \theta / 2} \begin{bmatrix} 1 & -\tan \theta / 2 \\ \tan \theta / 2 & 1 \end{bmatrix}$$

$$\therefore \text{Product} = \frac{1}{\sec^2 \theta / 2} \begin{bmatrix} 1 & -\tan \theta / 2 \\ \tan \theta / 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & -\tan \theta / 2 \\ \tan \theta / 2 & 1 \end{bmatrix}$$

$$= \frac{1}{\sec^2 \theta / 2} \begin{bmatrix} 1 - \tan^2 \theta / 2 & -2 \tan \theta / 2 \\ 2 \tan \theta / 2 & 1 - \tan^2 \theta / 2 \end{bmatrix}$$

$$= \begin{bmatrix} \cos^2 \theta / 2 & \sin^2 \theta / 2 & -2 \sin \theta / 2 \cos \theta / 2 \\ 2 \sin \theta / 2 & \cos \theta / 2 & \cos^2 \theta / 2 - \sin^2 \theta / 2 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

Illustration 32

If $A = \begin{bmatrix} 0 & -1 & 2 \\ 2 & -2 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}$ and $M = AB$, then M^{-1} is equal to

(A) $\begin{bmatrix} 2 & -2 \\ 2 & 1 \end{bmatrix}$

(B) $\begin{bmatrix} 1/3 & 1/3 \\ -1/3 & 1/6 \end{bmatrix}$

(C) $\begin{bmatrix} 1/3 & -1/3 \\ 1/3 & 1/6 \end{bmatrix}$

(D) $\begin{bmatrix} 1/3 & -1/3 \\ -1/3 & 1/6 \end{bmatrix}$

Ans. (C)

Solution:

$$M = \begin{bmatrix} 0 & -1 & 2 \\ 2 & -2 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ -2 & 2 \end{bmatrix}$$

$$\Rightarrow |M| = 6, \text{adj } M = \begin{bmatrix} 2 & -2 \\ 2 & 1 \end{bmatrix}$$

$$\therefore M^{-1} = \frac{1}{6} \begin{bmatrix} 2 & -2 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 1/3 & -1/3 \\ 1/3 & 1/6 \end{bmatrix}$$

SYSTEM OF EQUATION & CRITERIA FOR CONSISTENCY:

Gauss - Jordan Method:

$$a_1x + b_1y + c_1z = d_1$$

$$a_2x + b_2y + c_2z = d_2$$

$$a_3x + b_3y + c_3z = d_3$$

$$\Rightarrow \begin{bmatrix} a_1x + b_1y + c_1z \\ a_2x + b_2y + c_2z \\ a_3x + b_3y + c_3z \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix} \Rightarrow \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

$$\Rightarrow AX = B \quad \dots(i)$$

Multiplying $\text{adj } A$ on both the sides of (i)

$$\Rightarrow (\text{adj } A)AX = (\text{adj } A)B \Rightarrow |A|X = (\text{adj } A)B$$

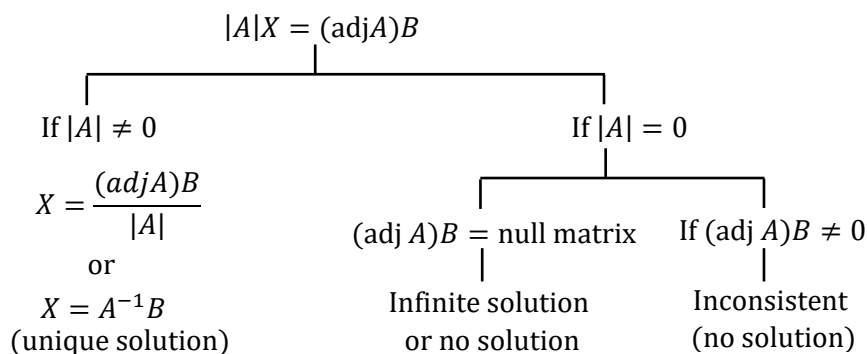


Illustration 33

$$x + y + z = 6$$

Solve the system $x - y + z = 2$ using matrix method.

$$2x + y - z = 1$$

Solution:

$$\text{Let } A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ \& } B = \begin{bmatrix} 6 \\ 2 \\ 1 \end{bmatrix}$$

Then the system is $AX = B$.

$\Delta = 6$, hence A is nonsingular,

$$\text{Cofactor } A = \begin{bmatrix} 0 & 3 & 3 \\ 2 & -3 & 1 \\ 2 & 0 & -2 \end{bmatrix}$$

$$\Rightarrow \text{adj } A = \begin{bmatrix} 0 & 2 & 2 \\ 3 & -3 & 0 \\ 3 & 1 & -2 \end{bmatrix}$$

$$\Rightarrow A^{-1} = \frac{1}{|A|} \text{adj } A = \frac{1}{6} \begin{bmatrix} 0 & 2 & 2 \\ 3 & -3 & 0 \\ 3 & 1 & -2 \end{bmatrix} = \begin{bmatrix} 0 & 1/3 & 1/3 \\ 1/2 & -1/2 & 0 \\ 1/2 & 1/6 & -1/3 \end{bmatrix}$$

$$\Rightarrow X = A^{-1} B = \begin{bmatrix} 0 & 1/3 & 1/3 \\ 1/2 & -1/2 & 0 \\ 1/2 & 1/6 & -1/3 \end{bmatrix} \begin{bmatrix} 6 \\ 2 \\ 1 \end{bmatrix} \text{ i.e. } \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$\Rightarrow x = 1, y = 2, z = 3$$

Illustration 34

The system of equations $x_1 - x_2 + x_3 = 2$, $3x_1 - x_2 + 2x_3 = -6$ and $3x_1 + x_2 + x_3 = -18$ has

- (A) No solution (B) Exactly one solution
(C) Infinite solutions (D) None of these

Ans. (C)

Solution:

$$D = \begin{vmatrix} 1 & -1 & 1 \\ 3 & -1 & 2 \\ 3 & 1 & 1 \end{vmatrix} = 1[-1-2] - 1[6-3] + 1[3+3] = 0$$

$$\text{And } D_1 = \begin{vmatrix} 2 & -1 & 1 \\ -6 & -1 & 2 \\ -18 & 1 & 1 \end{vmatrix} = 2(-1-2) - 1(-36+6) + 1(-6-18)$$

$$= -6 + 30 - 24 = 0$$

$$\text{Also, } D_2 = 0; D_3 = 0$$

So, the system is consistent ($D = D_1 = D_2 = D_3 = 0$)

i.e. system has infinite solution.

FINDING INVERSE USING ELEMENTARY ROW OPERATION:

Let X, A and B be matrices of, the same order such that $X = AB$. In order to apply a sequence of elementary row operations on the matrix equation $X = AB$, we will apply these row operations simultaneously on X and on the first matrix A of the product AB on RHS.

Similarly, in order to apply a sequence of elementary column operations on the matrix equation $X = AB$, we will apply, these operations simultaneously on X and on the second matrix B of the product AB on RHS.

In view of the above discussion, we conclude that if A is a matrix such that A^{-1} exists, then to find A^{-1} using elementary row operations, write $A = IA$ and apply a sequence of row operation on $A = IA$ till we get, $I = BA$. The matrix B will be the inverse of A . Similarly, if we wish to find A^{-1} using column operations, then, write $A = AI$ and apply a sequence of column operations on $A = AI$ till we get, $I = AB$.

Note:

In case, after applying one or more elementary row (column) operations on $A = IA$ ($A = AI$), if we obtain all zeros in one or more rows of the matrix A on L.H.S., then A^{-1} does not exist.

Illustration 35

Obtain the inverse of the following matrix using elementary operations $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$.

Solution:

$$\text{Write } A = IA, \text{ i.e., } \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 3 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \text{ (applying } R_1 \leftrightarrow R_2)$$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & -5 & -8 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & -3 & 1 \end{bmatrix} A \text{ (applying } R_3 \rightarrow R_3 - 3R_1)$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & -5 & -8 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & -3 & 1 \end{bmatrix} A \text{ (applying } R_1 \rightarrow R_1 - 2R_2)$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 0 \\ 1 & 0 & 0 \\ 5 & -3 & 1 \end{bmatrix} A \text{ (applying } R_3 \rightarrow R_3 + 5R_2)$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 0 \\ 1 & 0 & 0 \\ \frac{5}{2} & \frac{-3}{2} & \frac{1}{2} \end{bmatrix} A \text{ (applying } R_3 \rightarrow \frac{1}{2}R_3)$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{-1}{2} & \frac{1}{2} \\ 1 & 0 & 0 \\ \frac{5}{2} & \frac{-3}{2} & \frac{1}{2} \end{bmatrix} \quad A \text{ (applying } R_1 \rightarrow R_1 - R_3)$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{-1}{2} & \frac{1}{2} \\ -4 & 3 & -1 \\ \frac{5}{2} & \frac{-3}{2} & \frac{1}{2} \end{bmatrix} \quad A \text{ (applying } R_2 \rightarrow R_2 - R_3)$$

$$\text{Hence } A^{-1} = \begin{bmatrix} \frac{1}{2} & \frac{-1}{2} & \frac{1}{2} \\ -4 & 3 & -1 \\ \frac{5}{2} & \frac{-3}{2} & \frac{1}{2} \end{bmatrix}$$

JEE-ADVANCED (PART-1):**Illustration 36**

If $A = \begin{bmatrix} 0 & -\tan \frac{\alpha}{2} \\ \tan \frac{\alpha}{2} & 0 \end{bmatrix}$ and I is a unit matrix of order 2. Show that $I + A = (I - A) \cdot \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$

Solution:

We have, $I + A = \begin{bmatrix} 1 & -\tan \frac{\alpha}{2} \\ \tan \frac{\alpha}{2} & 1 \end{bmatrix}$ and $I - A = \begin{bmatrix} 1 & \tan \frac{\alpha}{2} \\ -\tan \frac{\alpha}{2} & 1 \end{bmatrix}$

Now, $(I - A) \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix} = \begin{bmatrix} 1 & -\tan \frac{\alpha}{2} \\ \tan \frac{\alpha}{2} & 1 \end{bmatrix}$

$$= \begin{bmatrix} 1 & \tan \frac{\alpha}{2} \\ -\tan \frac{\alpha}{2} & 1 \end{bmatrix} \begin{bmatrix} \frac{1 - \tan^2 \frac{\alpha}{2}}{1 + \tan^2 \frac{\alpha}{2}} & \frac{-2 \tan \frac{\alpha}{2}}{1 + \tan^2 \frac{\alpha}{2}} \\ \frac{2 \tan \frac{\alpha}{2}}{1 + \tan^2 \frac{\alpha}{2}} & \frac{1 - \tan^2 \frac{\alpha}{2}}{1 + \tan^2 \frac{\alpha}{2}} \end{bmatrix} = \begin{bmatrix} 1 & t \\ -t & 1 \end{bmatrix} \begin{bmatrix} \frac{1 - t^2}{1 + t^2} & -\frac{2t}{1 + t^2} \\ \frac{2t}{1 + t^2} & \frac{1 - t^2}{1 + t^2} \end{bmatrix}, \text{ where } t = \tan \frac{\alpha}{2}.$$

$$= \begin{bmatrix} \frac{1 - t^2 + 2t^2}{1 + t^2} & \frac{-2t + t - t^3}{1 + t^2} \\ \frac{-t + t^3 + 2t}{1 + t^2} & \frac{2t^2 + 1 - t^2}{1 + t^2} \end{bmatrix} = \begin{bmatrix} \frac{1 + t^2}{1 + t^2} & \frac{-t(1 + t^2)}{1 + t^2} \\ \frac{t(1 + t^2)}{1 + t^2} & \frac{1 + t^2}{1 + t^2} \end{bmatrix} = \begin{bmatrix} 1 & -t \\ t & 1 \end{bmatrix} = \begin{bmatrix} 1 & -\tan \frac{\alpha}{2} \\ \tan \frac{\alpha}{2} & 1 \end{bmatrix} = I + A$$

Illustration 37

Determine the matrix B and C with integral element such that $A = \begin{bmatrix} -1 & 0 \\ k & -2 \end{bmatrix} = B^3 + C^3$, where k is any integer.

(A) $B = A + I, C = -I$

(B) $C = A + I, B = -I$

(C) $B = A - I, C = -I$

(D) $B = A + I, C = +I$

Ans. (A)

Solution:

$$A^2 = \begin{bmatrix} -1 & 0 \\ k & -2 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ k & -2 \end{bmatrix} \Rightarrow A^2 = \begin{bmatrix} 1 & 0 \\ -3k & 4 \end{bmatrix}$$

$$\Rightarrow A^2 + 3A = \begin{bmatrix} 1 & 0 \\ -3k & 4 \end{bmatrix} + \begin{bmatrix} -3 & 0 \\ 3k & -6 \end{bmatrix} = \begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix}$$

$$\Rightarrow A^2 + 3A + 2I = \begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix} + 2 \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \mathbf{O}.$$

$$\therefore A^3 + 3A^2 + 2A = \mathbf{O}.$$

$$\Rightarrow (A + I)^3 - A = I^3 \Rightarrow A = (A + I)^3 + (-I)^3$$

$$\Rightarrow B^3 + C^3 \Rightarrow B = A + I \text{ and } C = -I$$

Illustration 38

Let $F(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$ then

(A) $[F(x)]^{-1} = F(x)$

(B) $[F(x)]^{-1} = -[F(-x)]$

(C) $[F(x)]^{-1} = -F(x)$

(D) $[F(x)]^{-1} = F(-x)$

Ans. (D)

Solution:

$$\text{We have, } F(x)F(-x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos x & \sin x & 0 \\ -\sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3$$

$$\Rightarrow F(-x) \text{ is the inverse of matrix } F(x) \text{ i.e. } [F(x)]^{-1} = F(-x).$$

JEE-ADVANCED (PART-2):

Illustration 39

If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ then prove that value of f & g satisfying the matrix equation $A^2 + fA + gI = \mathbf{O}$ are equal to

$$-t_r(A) \text{ and determinant of } A \text{ respectively. Given } a, b, c, d \text{ are non-zero reals and } I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}; O = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Solution:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad A^2 + fA + gI = 0$$

$$\Rightarrow A^2 = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{bmatrix} a^2 + bc & ab + bd \\ ca + dc & cd + d^2 \end{bmatrix}$$

$$\Rightarrow f = (A)$$

$$= -(a + d)$$

$$\Rightarrow g = (ad - bc)$$

$$\Rightarrow \begin{bmatrix} a^2 + bc & ab + bd \\ ca + dc & cd + d^2 \end{bmatrix} - (a + d) \begin{bmatrix} a & b \\ c & d \end{bmatrix} + (ad - bc) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Illustration 40

For the matrix $A = \begin{bmatrix} 4 & -4 & 5 \\ -2 & 3 & -3 \\ 3 & -3 & 4 \end{bmatrix}$ find A^{-2} .

Solution:

$$A^{-2} \times A^{-1} \times A^{-1}$$

$$\Rightarrow A = \begin{bmatrix} 4 & -4 & 5 \\ -2 & 3 & -3 \\ 3 & -3 & 4 \end{bmatrix}; [C_{ij}] = \begin{bmatrix} 3 & -1 & -3 \\ 1 & 1 & 0 \\ -3 & 2 & 4 \end{bmatrix}$$

$$\Rightarrow \text{adj } A = [C_{ij}]^T = \begin{bmatrix} 3 & 1 & 3 \\ -1 & 1 & 2 \\ -3 & 0 & 4 \end{bmatrix}$$

$$\Rightarrow |A| = \begin{vmatrix} 4 & -4 & 5 \\ -2 & 3 & -3 \\ 3 & -3 & 4 \end{vmatrix}$$

$$= 4(3) + 4 - 15$$

$$= 16 - 15 = 1$$

$$A^{-1} = \frac{\text{Adj}(A)}{|A|}$$

$$\Rightarrow A^{-1} = \begin{bmatrix} 3 & 1 & 3 \\ -1 & 1 & 2 \\ -3 & 0 & 4 \end{bmatrix}$$

$$\Rightarrow A^{-1}A^{-1} = \begin{bmatrix} 3 & 1 & 3 \\ -1 & 1 & 2 \\ -3 & 0 & 4 \end{bmatrix} \begin{bmatrix} 3 & 1 & 3 \\ -1 & 1 & 2 \\ -3 & 0 & 4 \end{bmatrix} = \begin{bmatrix} 17 & 4 & -19 \\ -10 & 0 & 13 \\ -21 & -3 & 25 \end{bmatrix}$$

Illustration 41

Given $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 4 & 1 \\ 2 & 3 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix}$ find P such that $BPA = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$.

Solution:

$$BPA = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix} P \begin{bmatrix} 1 & 1 & 1 \\ 2 & 4 & 1 \\ 2 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}_{2 \times 3}$$

order of P will be 2×3

$$\Rightarrow \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 2 & 4 & 1 \\ 2 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\Rightarrow P = \begin{bmatrix} -4 & 7 & -7 \\ 3 & -5 & 5 \end{bmatrix}$$

JEE-ADVANCED (PART-3):

Illustration 42

$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 4 \end{bmatrix}$, $I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $A^{-1} = \frac{1}{6}(A^2 + cA + dI)$, then the value of c and d are -

(A) -6, -11

(B) 6, 11

(C) -6, 11

(D) 6, -11

Ans. (C)

Solution:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 4 \end{bmatrix} \Rightarrow |A| = 6$$

$$\Rightarrow A^{-1} \Rightarrow \frac{\text{adj}A}{|A|} = \frac{1}{6} \begin{bmatrix} 6 & 0 & 0 \\ 0 & 4 & -1 \\ 0 & 2 & 1 \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 5 \\ 0 & -10 & 14 \end{bmatrix}$$

$$\Rightarrow A^{-1} = \frac{1}{6} [A^2 + cA + dI]$$

$$\Rightarrow \frac{1}{6} \begin{bmatrix} 6 & 0 & 0 \\ 0 & 4 & -1 \\ 0 & 2 & 1 \end{bmatrix} = \frac{1}{6} \left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 5 \\ 0 & -10 & 14 \end{bmatrix} + \begin{bmatrix} c & 0 & 0 \\ 0 & c & c \\ 0 & -2c & 4c \end{bmatrix} + \begin{bmatrix} d & 0 & 0 \\ 0 & d & 0 \\ 0 & 0 & d \end{bmatrix} \right\}$$

on comparing we get

$$\Rightarrow -1 = 5 + c \Rightarrow c = -6$$

$$\Rightarrow 1 = 14 + 4c + d \Rightarrow 1 = 14 - 24 + d$$

$$\Rightarrow d = 11$$

Illustration 43

If $P = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$, $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $Q = PAP^T$ and $x = P^T Q^{2005} P$, then x is equal to -

(A) $\begin{bmatrix} 1 & 2005 \\ 0 & 1 \end{bmatrix}$

(B) $\begin{bmatrix} 4+2005\sqrt{3} & 6015 \\ 2005 & 4-2005\sqrt{3} \end{bmatrix}$

(C) $\frac{1}{4} \begin{bmatrix} 2+\sqrt{3} & 1 \\ -1 & 2-\sqrt{3} \end{bmatrix}$

(D) $\frac{1}{4} \begin{bmatrix} 2005 & 2-\sqrt{3} \\ 2+\sqrt{3} & 2005 \end{bmatrix}$

Ans. (A)

Solution:

$$PP^T = I$$

$$\Rightarrow A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \Rightarrow A^2 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

and so, on

$$\Rightarrow Q = PAP^T$$

$$\Rightarrow Q^2 = (PAP^T)(PAP^T) = PA^2P^T$$

$$\Rightarrow Q^{2005} = PA^{2005}P^T$$

$$\Rightarrow x = P^T(PA^{2005}P^T)P$$

$$\Rightarrow x = A^{2005} = \begin{bmatrix} 1 & 2005 \\ 0 & 1 \end{bmatrix}$$

Illustration 44

Let p be an odd prime number and T_p be the following set of 2×2 matrices:

$$T_p = \left\{ A = \begin{bmatrix} a & b \\ c & a \end{bmatrix} : a, b, c \in \{0, 1, 2, \dots, p-1\} \right\}$$

The number of A in T_p such that A is either symmetric or skew-symmetric or both, and $\det(A)$ divisible by p is -

(A) $(p-1)^2$

(B) $2(p-1)$

(C) $(p-1)^2 + 1$

(D) $2p-1$

Ans. (D)

Solution:

If A is symmetric, $A^T = A$

$$\Rightarrow \begin{bmatrix} a & b \\ c & a \end{bmatrix} = \begin{bmatrix} a & c \\ b & a \end{bmatrix} \Rightarrow b = c$$

If A is skew symmetric, $A^T = -A$

$$\Rightarrow \begin{bmatrix} a & b \\ c & a \end{bmatrix} = \begin{bmatrix} -a & -c \\ -b & -a \end{bmatrix} \Rightarrow a = 0, b + c = 0$$

$$\therefore b, c \geq 0 \Rightarrow a = 0, b = 0, c = 0$$

$$\text{Now, } \det(A) = a^2 - bc$$

$$= a^2 - b^2 \quad (\because b = c \text{ for } A \text{ being symmetric or skew symmetric or both})$$

$$= (a - b)(a + b) \text{ is divisible by } p.$$

$$\text{Let } (a - b)(a + b) = \lambda p, \lambda \in I$$

Range of $(a + b)$ is 0 to $2p - 2$ which includes only one multiple of p i.e. p

$$\therefore a + b = p \text{ and } a - b \in I$$

$$\Rightarrow \text{possible number of pairs of } a \text{ and } b \text{ will be } p - 1.$$

Also, range of $(a - b)$ is $1 - p$ to $p - 1$ which includes only one multiple of p i.e. 0

$$\therefore a - b = 0 \text{ and } a + b \in I$$

$$\Rightarrow \text{Possible number of pairs of } a \text{ and } b \text{ will be } p.$$

$$\text{Hence total number of } A \text{ in } T_p \text{ will be } p + p - 1 = 2p - 1.$$

JEE-ADVANCED (PART-4):

For illustration 45 to 47

Let A is matrix of order 2×2 such that $A^2 = O$.

Illustration 45

$A^2 - (a + d)A + (ad - bc)I$ is equal to

(A) I

(B) O

(C) $-I$

(D) none of these

Ans. (B)

Solution:

Let

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\Rightarrow A^2 - (a + d)A + (ad - bc)I$$

$$= \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} - (a + d) \begin{bmatrix} a & b \\ c & d \end{bmatrix} + (ad - bc) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} a^2 + bc & ab + bd \\ ac + cd & bc + d^2 \end{bmatrix} - \begin{bmatrix} a^2 + ad & ab + bd \\ ac + cd & ad + d^2 \end{bmatrix} + \begin{bmatrix} ad - bc & 0 \\ 0 & ad - bc \end{bmatrix} = O$$

Illustration 46

$\text{tr}(A)$ is equal to

- (A) 1 (B) 0 (C) -1 (D) none of these

Ans. (B)

Solution:

If $A = O$, $\text{tr}(A) = 0$. suppose $A \neq O$ and $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$$\Rightarrow |A| = 0 \text{ and } A^2 - (a + d)A + (ad - bc)I = 0$$

$$\Rightarrow a + d = 0$$

Illustration 47

$(I + A)^{100} =$

- (A) $100A$ (B) $100(I + A)$ (C) $100I + A$ (D) $I + 100A$

Ans. (D)

Solution:

$$(I + A)^{100} = {}^{100}C_0 I^{100} + {}^{100}C_1 I^{99}A + {}^{100}C_2 I^{98}A^2 + \dots + {}^{100}C_{100} A^{100}$$

$$= I + 100A + O + O + \dots + O$$

$$= I + 100A$$

For illustration 48 to 50

If A and B are two square matrices of order 3×3 which satisfy $AB = A$ and $BA = B$

Illustration 48

Which of the following is true?

- (A) If matrix A is singular then matrix B is non-singular
 (B) If matrix A is non-singular then matrix B is singular
 (C) If matrix A is singular then matrix B is also singular
 (D) Cannot say anything.

Ans. (C)

Solution:

$$AB = A \Rightarrow |AB| = |A| \quad \dots(1)$$

$$\Rightarrow |A| = 0 \text{ or } |B| = 1$$

$$\Rightarrow BA = B \Rightarrow |BA| = |B| \quad \dots(2)$$

$$\Rightarrow |A| = 1 \text{ or } |B| = 0$$

If $|A| = 0$, then from Eq. (2), $|B| = 0$

If $|B| = 0$, then from Eq. (1), $|A| = 0$

Illustration 49

$(A + B)^9$ is equal to

- (A) $9(A + B)$ (B) $9 \cdot I_{3 \times 3}$ (C) $256(A + B)$ (D) $256 I$

Ans. (C)

Solution:

$$AB = A, BA = B$$

$$\Rightarrow ABA = A^2 \Rightarrow A(BA) = A^2 \Rightarrow AB = A^2 \Rightarrow A = A^2$$

$$\text{Similarly, } B^2 = B$$

$$\Rightarrow (A + B)^2 = A^2 + B^2 + AB + BA$$

$$\Rightarrow (A + B)^2 = A + B + A + B = 2(A + B)$$

$$\Rightarrow (A + B)^3 = (A + B)^2(A + B) = 2(A + B)^2 = 2^2(A + B)$$

$$\Rightarrow (A + B)^9 = 2^8(A + B) = 256(A + B)$$

Illustration 50

$(A + I)^5$ is equal to (where I is identity matrix)

(A) $I + 60I$

(B) $I + 16A$

(C) $I + 31A$

(D) none of these

Solution:

Ans. (C)

$$(A + I)^5 = I + 5A + 10A^2 + 10A^3 + 5A^4 + A^5$$

$$= I + 5A + 10A + 10A + 5A + A$$

$$(\because A^2 = A \Rightarrow A^3 = A^4 = A^5 = \dots = A)$$

$$= I + 31A$$